Experiment – 8 : User Management and Role-Based Access Control

Aim

To implement user management and role-based access control in the Event Desk event management platform using Zoho Creator, enabling secure user administration, controlled access to application features, and appropriate permissions for different categories of users.

Theory

User Management and Role-Based Access Control (RBAC) are essential security mechanisms used to manage application users and control access to data and functionality. In an event management platform, different users require different levels of access depending on their responsibilities. For example, an event administrator may need to manage participant registrations and monitor analytics, while a participant may only need access to their own registration information.

Zoho Creator provides user management, application permissions, roles, and access-control settings to help administrators manage application access. By assigning suitable permissions to users, the platform can restrict access to sensitive information and prevent unauthorized operations.

The following features are relevant to the Event Desk platform:

1. User Management

The User Management form maintains information about users associated with the platform. It provides a structured way to organize user details and support administrative management.

2. Role-Based Access Control

RBAC assigns permissions according to a user's role rather than granting every user identical access. This helps ensure that users can access only the functions required for their responsibilities.

3. Administrator Access

Administrators require access to registration records, user management, reports, workflows, and analytics dashboards. Administrative permissions allow authorized users to oversee registration activities and manage the application.

4. Participant Access

Participants require access to relevant registration functions and their own registration information. Restricting participant access helps protect other users' personal details and prevents unauthorized administrative actions.

5. Permission Management

Permissions determine whether users can view, add, edit, or delete records and access specific application components. Appropriate permission settings help maintain data integrity and reduce the risk of unauthorized changes.

6. Data Privacy and Security

Event registration records may contain names, email addresses, phone numbers, and payment-related status information. Access restrictions help protect this information from unauthorized viewing or modification.

7. Secure Registration Management

Role-based permissions help protect participant records, registration statuses, reports, and administrative workflows. They also support the separation of administrative responsibilities from participant activities.

8. Principle of Least Privilege

The principle of least privilege recommends granting users only the permissions necessary to perform their tasks. Applying this principle reduces unnecessary access and improves the overall security of the platform.

9. Access Validation

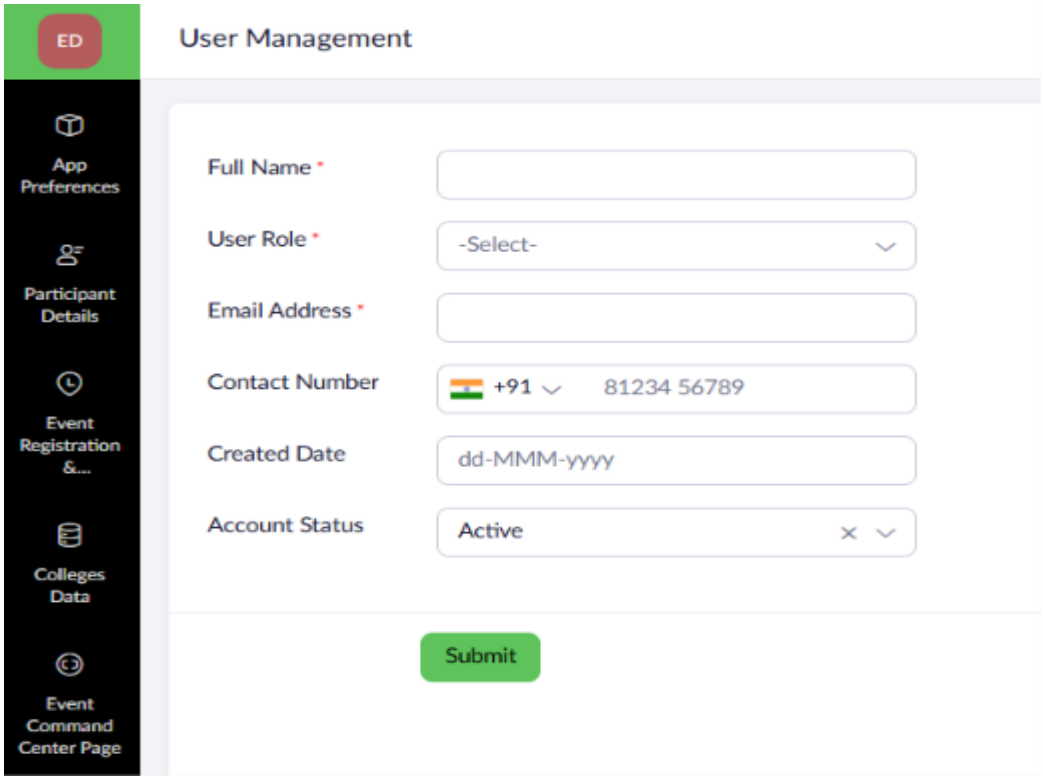
Access control should be evaluated by checking whether users with different roles can perform permitted actions and whether restricted actions are denied. This helps identify incorrect permission settings and potential security gaps.

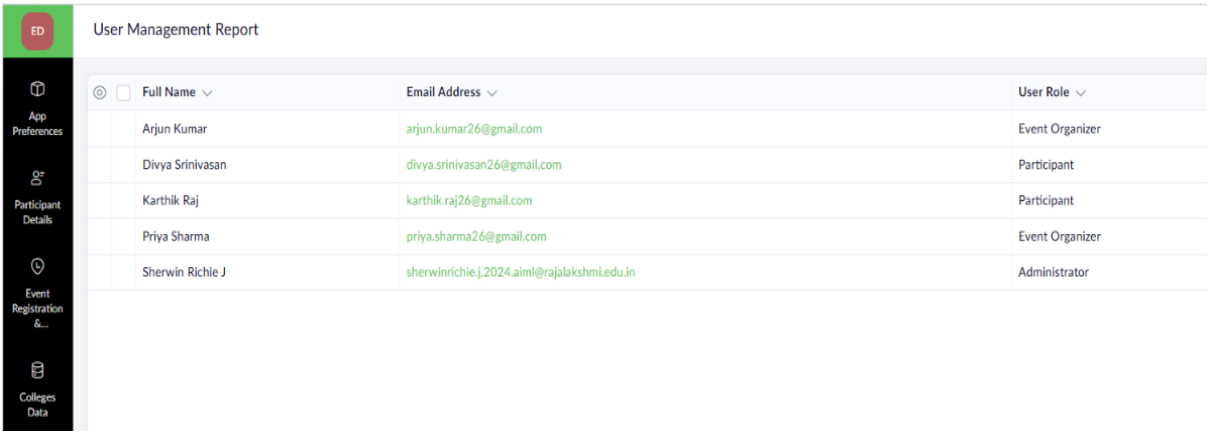
Procedure

1. Open the Event Desk application in Zoho Creator.
2. Locate the existing User Management form and examine the fields used to maintain user information.

3. Identify the different categories of users who require access to the event management platform.
4. Define the responsibilities of each user category, such as administrators and participants.
5. Open the application's user, role, and permission settings in Zoho Creator.
6. Configure or review the available roles according to the responsibilities of each user category.
7. Assign appropriate permissions to administrators for managing registration records, reports, dashboards, and other administrative functions.
8. Configure participant permissions to provide access to relevant registration functions while restricting access to other participants' information.
9. Review permissions for viewing, adding, editing, and deleting records to ensure that each role has only the necessary access.
10. Examine access to the Event Registration Form, Event Registration Report, Registration Status Board, Participation Dashboard, and AI Registration Analytics Dashboard.
11. Ensure that sensitive participant information is accessible only to authorized users and that administrative functions are restricted appropriately.
12. Check the application's authentication and authorization settings to ensure that users must have appropriate access before using protected features.
13. Test the configured permissions using suitable test accounts or authorized test users assigned to different roles.
14. Verify that each role can perform its permitted operations and that restricted operations are not accessible.
15. Review the permission settings and correct any inconsistencies identified during access validation.

Screenshots





Result

The User Management and Role-Based Access Control design for the Event Desk event management platform establishes a structured approach to managing users and controlling access to application features. The User Management form supports organized user administration, while role-based permissions help separate administrative functions from participant activities.